Name: **Enrolment No:**	MANIPAL UNIVERSITY JAIPUR

Odd Semester Mid Term Examination, November 2024
Faculty of Engineering, School of Computer Science and Engineering
Department of Computer Science and Engineering
BTech - CSE

Course Code: CSE2103	**Course: Computer Organization and Architecture**	**Semester: III**
Time: 1.5 hrs.	**Solution**	**Max. Marks: 30**

Instructions: All questions are compulsory.
Missing data, if any, may be assumed suitably.
Calculator is allowed.

SECTION A

S.No.		Marks	CO
Q A1	Write the steps needed to execute the machine instruction *Load LOCA, R0* in terms of transfers between the functional components of computer system (Memory and Processor) and some simple control commands. Assume that the instruction itself is stored in the memory at location INSTR and that this address is initially in register PC. • Transfer the contents of register PC to register MAR • Issue a Read command to memory, and then wait until it has transferred the requested word into register MDR • Transfer the instruction from MDR into IR and decode it • Transfer the address LOCA from IR to MAR • Issue a Read command and wait until MDR is loaded • Transfer contents of MDR to R0 • Transfer contents of PC to ALU • Add 1 to operand in ALU and transfer incremented address to PC	[2]	CO1
Q A2	What are the main pros and cons of associative and direct mapping, respectively? Associative mapping Pros- efficient cache utilization Cons - costlier to implement (all cache lines or tags need to be checked for each memory reference) Direct mapping Pros – Simple to implement (only one cache line or tag need to be checked for each memory reference)	[2]	CO3

	Cons – not flexible (efficient cache utilization is not there)		
Q A3	What is the main advantage of Sequential multiplier over Array multiplier? • Less gates are required • Can be used for signed multiplication	[2]	CO4
SECTION B			
Q B1	Suppose Machine A executes a program with an average CPI (Cycles Per Instruction) of 2.3. Machine B, which uses the same instruction set but has a more efficient compiler, executes the same program with 20% fewer instructions and an average CPI of 1.7, operating at a clock rate of 1.2 GHz. What should be the clock rate of Machine A so that it achieves the same performance as Machine B. To ensure that Machine A and Machine B have the same performance, we use the formula for execution time: $$(\text{Instruction Count}) \times (\text{CPI}) \times (\text{Clock Cycle Time}) = \text{Execution Time}$$ For both machines to have the same execution time, we equate the expressions for Machine A and Machine B: $$\text{ICA} \times 2.3 \times \text{CA} = 0.80 \times \text{ICA} \times 1.7 \times \left(\frac{1}{1.2 \times 10^9}\right)$$ Simplifying, we solve for CA (the clock cycle time of Machine A): $$\text{CA} = 0.49 \times 10^{-9} \text{ sec}$$ Thus, the clock rate of Machine A is: $$\text{Clock Rate of A} = \frac{1}{\text{CA}} = 2.04 \text{ GHz}$$	[4]	CO1
Q B2	Consider a hypothetical processor which has the following addressing modes as given below along with their frequency of occurrence. Addressing Mode — Frequency (%) Register — 20 Immediate — 20 Direct — 30 Register indirect — 15 Indexed — 15	[4]	CO2

<table>
<tr>
<td></td>
<td>Assume that 5 clock cycles are required for memory reference, 1 clock cycle is required for register reference, 1 clock cycle is required for arithmetic computation and zero clock cycles when the operand is specified in the instruction itself. Determine the average number of clock cycles required for operand fetch on this processor.

Avg clock cycles = (20 × 1 + 20 × 0 + 30 × 5 + 15 × 6 + 15 × 7) / 100
= (20 + 150 + 90 + 105) / 100 = 36.5</td>
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<tr>
<td>Q B3</td>
<td>Consider an 8-way set associative mapped cache. The size of the main memory is 1 GB and there are 10 bits in the tag. Compute the size of cache memory.

Tag | Set | offset.
10 | y. | x
30 bits.

Total sets in cache = 2^y
" Blocks " = 2^{y+3}
size of cache = $2^{y+3} \times 2^x = 2^{x+y+3}$
= 2^{23} bytes = 8 MB</td>
<td>[4]</td>
<td>CO3</td>
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<tr>
<td>Q B4</td>
<td>Convert the Multiplicand M = $(-1)_{10}$ and Multiplier Q = $(-112)_{10}$ represented as decimal numbers, in 8-bit, signed, 2's complement binary numbers. Use Bit-pair multiplication methods to multiply them.

1 1 1 1 1 1 1 1 (-1)
1 0 0 1 0 0 0 0 (-112)
-1 0 +1 -1 0 0 0 0 ← Booth encoding.
-2 +1 0 0 ← Bit pair encoding

0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0
1 1 1 1 1 1 1 1
0 0 0 0 1 0</td>
<td>[4]</td>
<td>CO4</td>
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0 0 0 0 0 1 1 1 0 0 0 0 (112)

<table>
<tr><th colspan="4">SECTION-C</th></tr>
<tr><td>Q C1</td><td>Consider byte addressable machine with 32-bit word size. An integer array of order m × n is stored starting from memory location LIST in column major order (that is, the elements of array are stored column-wise at successive memory location starting from LIST). Write an assembly code for computing the sum of integers of each row of the array and store these sums in the memory word locations at addresses SUM, SUM+4, SUM +8…. (. i.e., m successive locations starting from SUM). Assume that values of m and n are stored at memory location M and N respectively.
Ans
<pre>
         Move M, R4
         Multiply #4, R4
         Move #LIST, R1
         Move #SUM, R3
         Move M, R10
OUTER   Move N, R11
         Clear R2
         Clear R0
INNER   Add (R1, R2), R0
         Add R4, R2
         Decrement R11
         Branch>0 INNER
         Move R0, (R3)
         Add #4, R3
         Add #4, R1
         Decrement R10
Branch>0 OUTER
</pre></td><td>[5]</td><td>CO2</td></tr>
<tr><td>Q C2</td><td>Calculate gate delays required to produce C_{48} and S_{47} for the following 48-bit adders.
a) 48-bit ripple carry adder.
b) 48-bit adder by cascading 4-bit CLA (Carry Look Ahead) adders. Here, 4-bit CLA are developed using first level Generate and Propagate functions.
Ans
a) For Carry $C_{48} = 2 * 48 = 96$ gate delays (GD)
For Sum $S_{47} = 95$ GD
b) For Carry $C_{48} = 3 + 11 * 2 = 25$ GD
For Sum $S_{47} = 26$ GD</td><td>[3]</td><td>CO4</td></tr>
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